1. Attention helps a word use its context

A word alone may be unclear. Other words help us understand it.

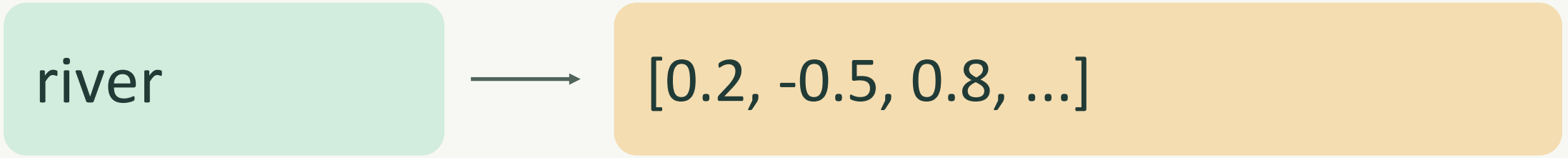
We sat on the river bank.

I put money in the bank.

Attention mixes information from other allowed positions.

2. An embedding represents a token with numbers

Computers work with numbers. An embedding is a list of numbers for a token.



Illustrative numbers, not actual learned river features.

Embedding = a numerical representation. Position = where the token occurs.

3. A query is the vector used to look for a match

Think: which information could help update this position?

bank
The position being updated

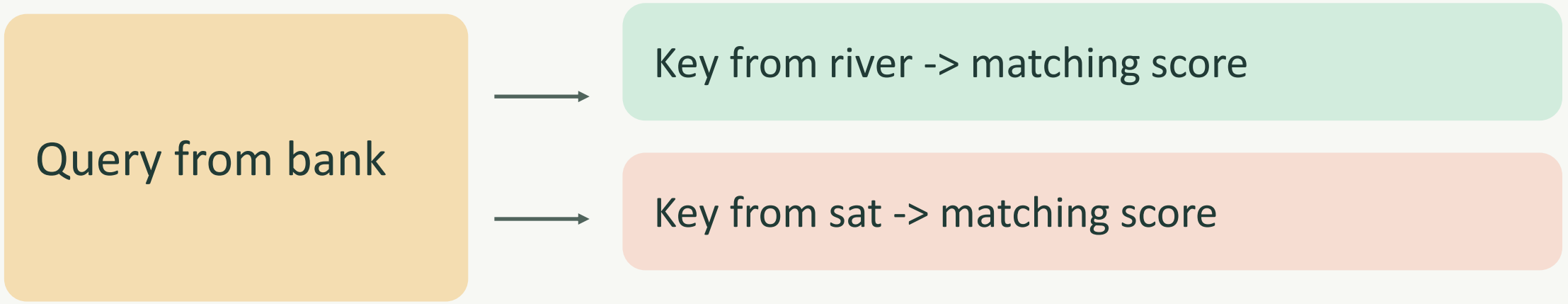
→

Query
A vector for matching useful context

"What would help here?" is an analogy, not the actual query vector.

4. A key is the vector used to be matched

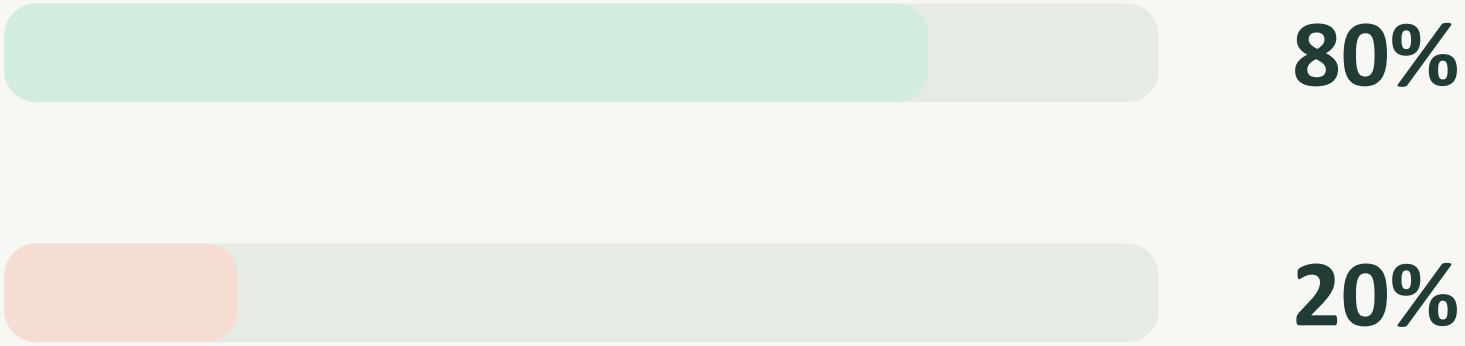
The query is compared with a key from each available position.



Query + key -> a score. A score is not the information we retrieve.

5. Attention weights are mixing percentages

Scores become weights. For one query, the weights add up to 100%.



Illustrative weights: 80% + 20% = 100%. Softmax produces the proportions.

6. A value is the information being mixed

The output is a blend of value vectors. It is not a new word.

80% of [10, 0] = [8, 0]

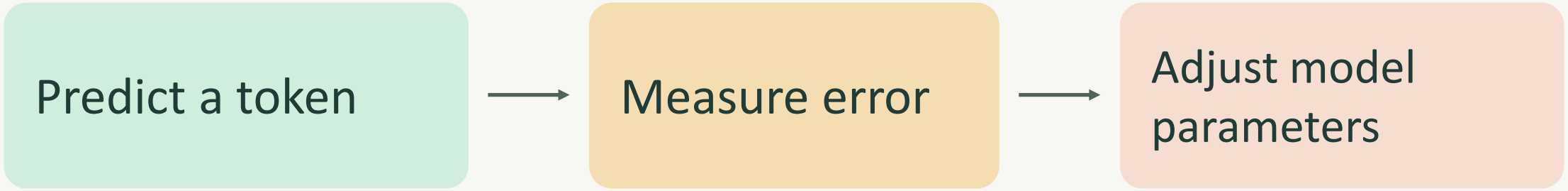
20% of [0, 10] = [0, 2]

Add them -> output [8, 2]

Key = matching features. Value = content. Weight = amount. Output = blend.

7. The model learns the transformations

Training learns how to create queries, keys and values from representations.



Example training text: "I drink a cup of ___" -> target: "tea"

Training learns the rules. A sentence supplies the input to those rules.

8. A text generator must not peek ahead

Self-attention reads the same sequence. Causal masking hides future tokens.

I

like

coffee <- hidden target

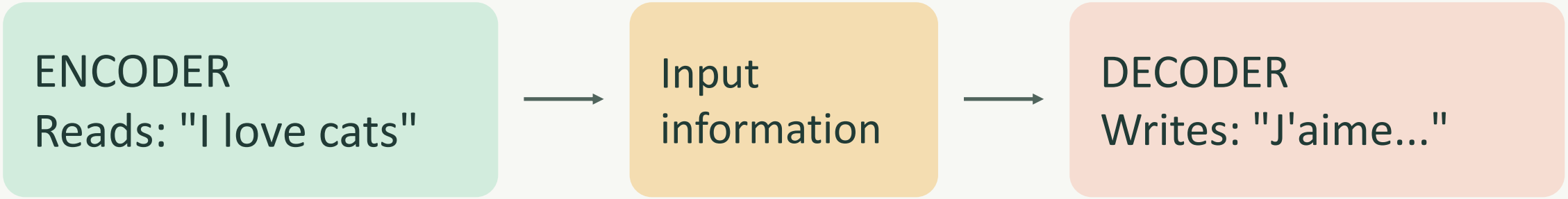
Allowed context: "I like"

Predict next: "coffee"

Mask = an access rule. Read the available context, not the future answer.

9. Encoder reads. Decoder writes.

These are two roles. An encoder-decoder model connects them.



Cross-attention = the decoder consulting the encoder.

10. A Transformer combines a few jobs

Attention shares information. Other components process and preserve it.

Position

Order matters: dog bites man

Multiple heads

Several learned mixtures at once

Feed-forward

Process features at each position

Residual

Old + update: [2,1] + [0,3]

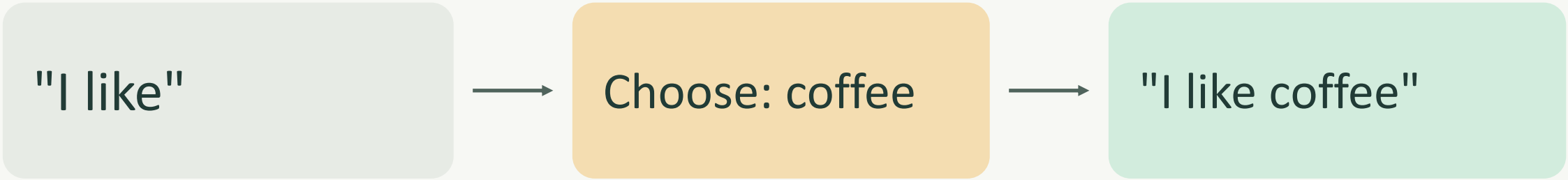
Normalization

Rescale features to regulate magnitudes

One block updates representations. Stacked blocks repeat the process.

11. GPT predicts one more token, then repeats

After the blocks, a vocabulary projection produces next-token scores.



Then predict again: "I like coffee."

Attention output is internal information. Token selection happens later.

12. The whole idea in three sentences

Match information. Mix it. Use the result to update a representation.

Compare query with keys -> matching scores

Mix values using weights -> a blended vector

Transformer blocks -> updated representations